

Bahir Dar Institute of Technology

Faculty of Electrical and Computer Engineering

CHAPTER TWO

BASIC LAWS

OHM'S LAW

- Relates the three basic electrical quantities(i.e. Voltage, current & resistance) mathematically.
- Ohm's law states that current is directly proportional to voltage and inversely proportional to resistance, which is given in the following formula:

$$I = V/R.....*$$

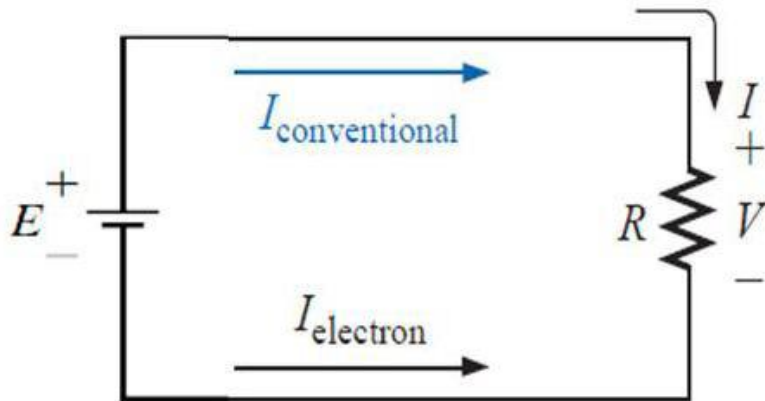
- For a constant value of R, if the value of V is increased, the value of I increases; if V is decreased, I decreases. If V is constant and R is increased, I decreases. Similarly, if V is constant and R is decreased, I increases.
- By manipulating Equation *, you can obtain expressions for voltage and resistance.

$$V = IR \text{ and } R = V/I$$

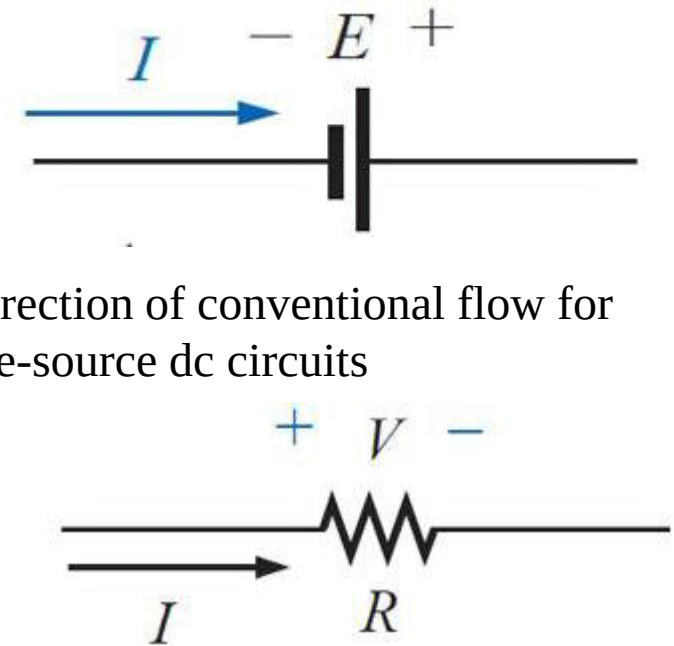
ELECTRICAL CIRCUITS AND INTRODUCTORY DC CIRCUIT ANALYSIS

- An electrical Ckt is an interconnection of electrical components.
- A simple & basic electrical circuit consists of:
 - Voltage source(E)
 - The load; is a device that dissipate/ converts electrical energy.
 - The connecting wires
- The voltage source pressures charges to flow through the circuit.

Consider the following Circuit



Defining the direction of conventional flow for single-source dc circuits



Note:

- By convention, the direction of conventional current flow is opposite to that of electron flow.
- Conventional current flow always passes from a high(+ve) to a low(-ve) potential when passing through a resistor for any number of voltage sources in the same circuit.
- By following the direction of conventional current flow, we notice that there is a rise in potential across the battery (-ve to +ve), and a drop in potential across the resistor (+ve to -ve)

❑ Depending on how the components are connected, electrical circuits can be grouped in to four types:

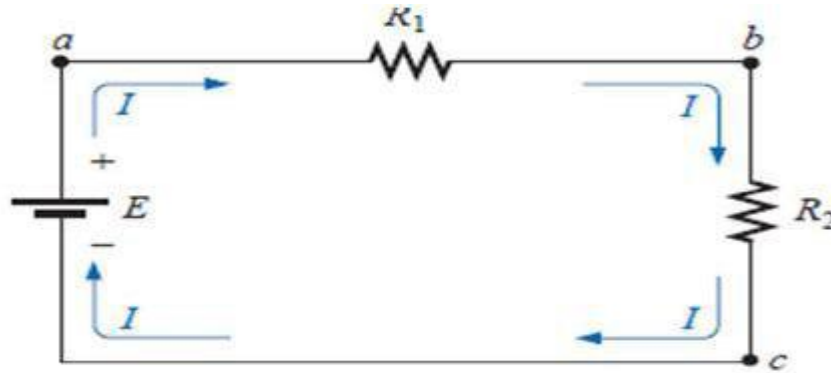
- 1) Series Circuit
- 2) Parallel Circuit
- 3) Series-Parallel Circuit
- 4) Complex Circuit

1. SERIES CIRCUIT

➤ Two elements are in series if:

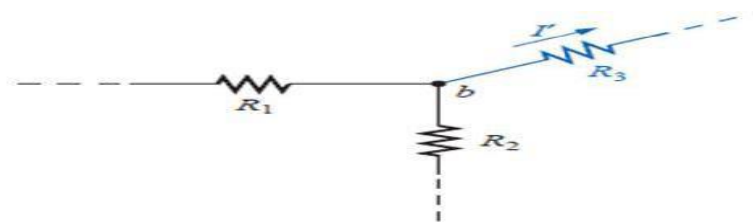
- They have only one terminal in common (i.e., one lead of one is connected to only one lead of the other).
- The common point between the two elements is not connected to another current-carrying element

- The circuit of Fig. (a) has three elements joined at three terminal points (a, b, and c) to provide a closed path for the current I . Since all the elements are in series, the network is called a series circuit.



(a) Series circuit

- Current will leave the positive terminal of the voltage source, move through the resistors, and return to the negative terminal of the source.
- If the above circuit is modified such that a current-carrying resistor R_3 is introduced, as shown in Fig.(b) below, the resistors R_1 and R_2 are no longer in series due to a violation of number 2 of the above definition of series elements.



(b) R_1 and R_2 are not in series

TOTAL CURRENT IN SERIES CKT(IT)

- A series ckt provides only one path for current between two points in a circuit, so that the current is the same through series elements. i.e

$$I_S = I_1 = I_2 = I_3 = \dots = I_n$$

TOTAL RESISTANCE IN SERIES CKT(RT)

- The total resistance of a series circuit is the sum of the resistance levels.
- In general, to find the total resistance of N resistors in series, the following equation is applied:

$$R_T = R_1 + R_2 + R_3 + \dots + R_N \quad (\text{ohms, } \Omega)$$

SOURCE CURRENT(IS)

Once R_T is known, the current drawn from the source can be determined using Ohm's law, as follows

$$I_s = \frac{E}{R_T} \quad (\text{amperes, A})$$

VOLTAGE DROP ACROSS EACH RESISTOR

□ The fact that the current is the same through each element of series circuit, the voltage across each resistor can be calculated using Ohm's law:

$$V_1 = IR_1, V_2 = IR_2, V_3 = IR_3, \dots, V_N = IR_N$$

From this, the total sum of each voltage drops across each resistor is equal to the source voltage(E).

$$E = V_1 + V_2 + V_3 + \dots + V_n$$

The power delivered to each resistor can then be determined using any one of three equations as listed below for R1:

$$P_1 = V_1 I_1 = I_1^2 R_1 = \frac{V_1^2}{R_1}$$

Generally, delivered power at each resistor can be calculated

$$P_n = V_n I = \frac{V_n^2}{R_n} = I^2 R_n$$

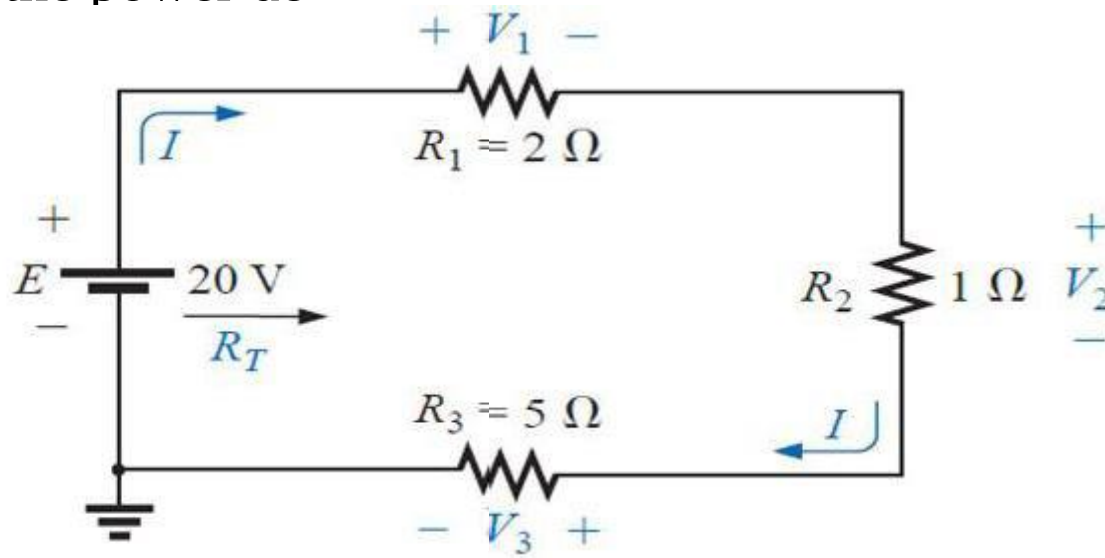
- The power delivered by the source is: **$P_{del} = EI$**
- The total power delivered to a resistive circuit is equal to the total power dissipated by the resistive elements.

That is:

$$P_{del} = P_1 + P_2 + P_3 + \dots + P_n$$

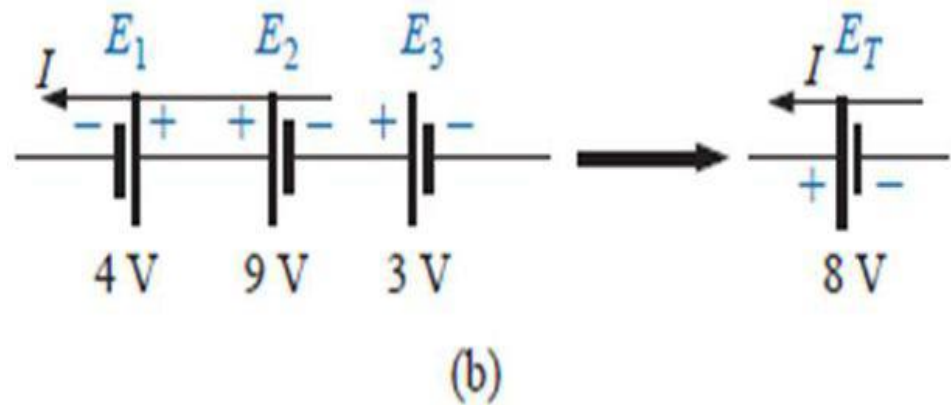
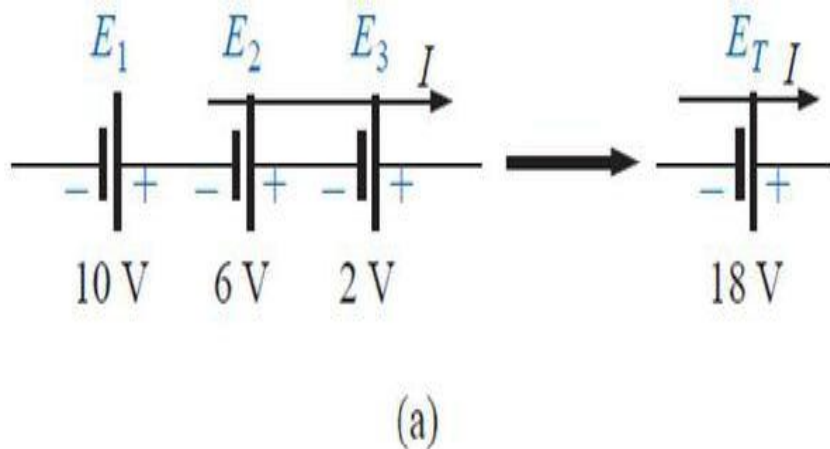
EXAMPLE

- Find the total resistance for the following series circuit
- Calculate the source current I_s .
- Determine the voltages V_1 , V_2 , and V_3 .
- Calculate the power dissipated by R_1 , R_2 , and R_3 .
- Determine the power de



VOLTAGE SOURCES IN SERIES

- Voltage sources can be connected in series, as shown in following Fig. (a) and Fig. (b), to increase or decrease the total voltage applied to a system.
- The total Voltage is equal to the algebraic sum of the individual Voltages, which are connected in series.
- The net Polarity is the polarity of the larger sum and when summing Voltages, take care of their polarities, Example :-20



KIRCHHOFF'S VOLTAGE LAW(KVL)

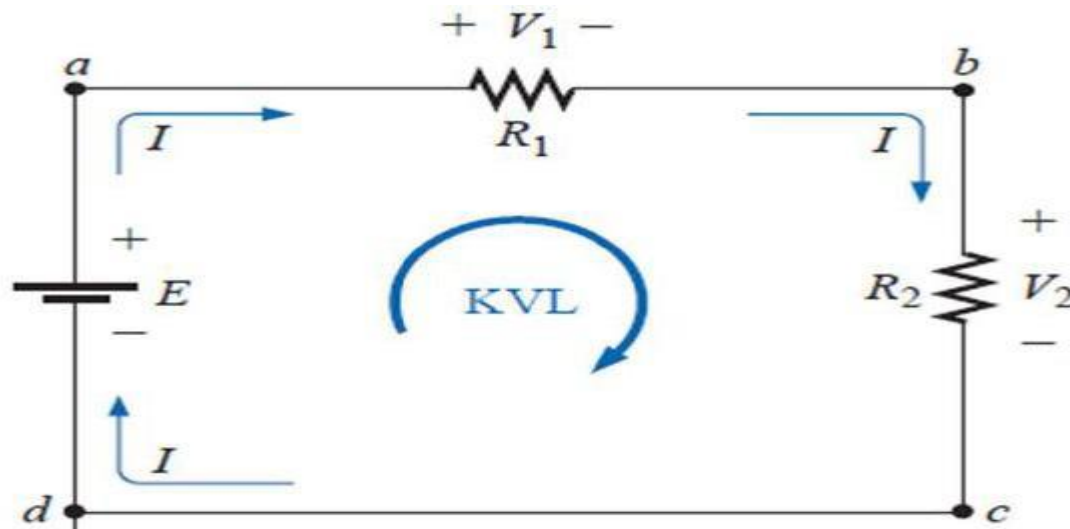
- Provides an important relationship among voltage levels around any closed loop of a network.
- KVL states that “the algebraic sum of the potential rises and drops around a closed loop (or path) is zero.”

That is:

$$\sum_{\text{loop}} V = 0$$

(Kirchhoff's voltage law
in symbolic form)

N.B: A closed loop is any continuous path that leaves a point in one direction and returns to that same point from another direction without leaving the circuit, Consider the following series ckt



To apply KVL:

- Use either CW or CCW direction,(CW direction is chosen for uniformity)
- A plus sign is assigned to a potential rise (- to +), and a minus sign to a potential drop (+ to -).

$$+E - V_1 - V_2 = 0$$

$$E = V_1 + V_2$$

Applying KVL, to the circuit above, we have:

Implies, the applied voltage of a series circuit equals the sum of the voltage drops across the series elements.

KVL can be stated in another form as:

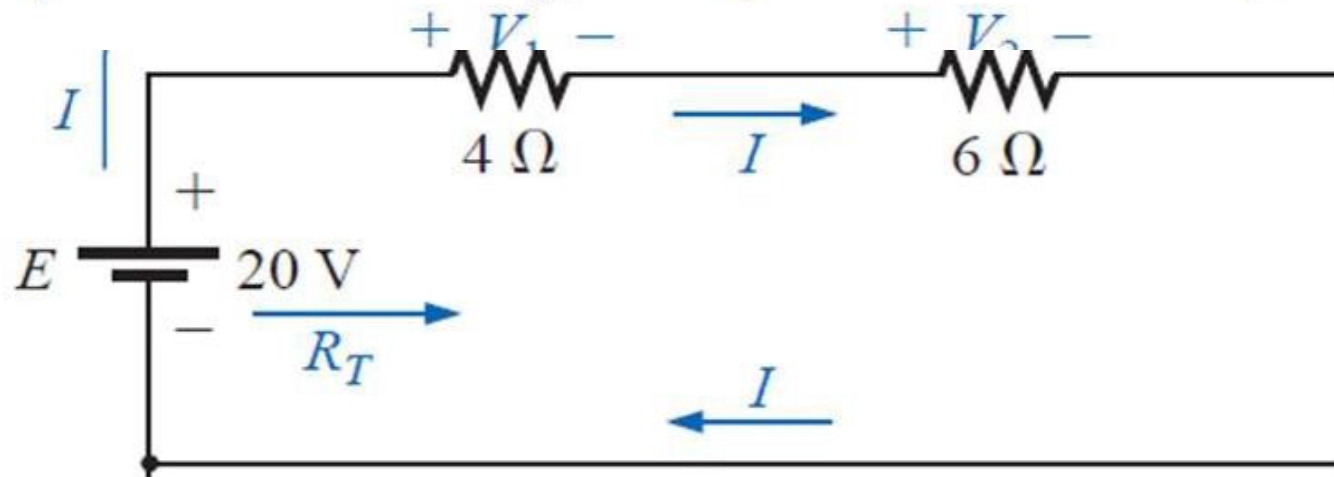
$$\sum_{\curvearrowright} V_{\text{rises}} = \sum_{\curvearrowright} V_{\text{drops}}$$

Which in words states that the sum of the rises around a closed loop must equal the sum of the drops in potential.

Examples

1) for the following circuit:

- Find R_T .
- Find I .
- Find V_1 and V_2 .
- Find the power to the $4\text{-}\Omega$ and $6\text{-}\Omega$ resistors.
- Find the power delivered by the battery, and compare it to that dissipated by the $4\text{-}\Omega$ and $6\text{-}\Omega$ resistors combined.
- Verify Kirchhoff's voltage law (clockwise direction).



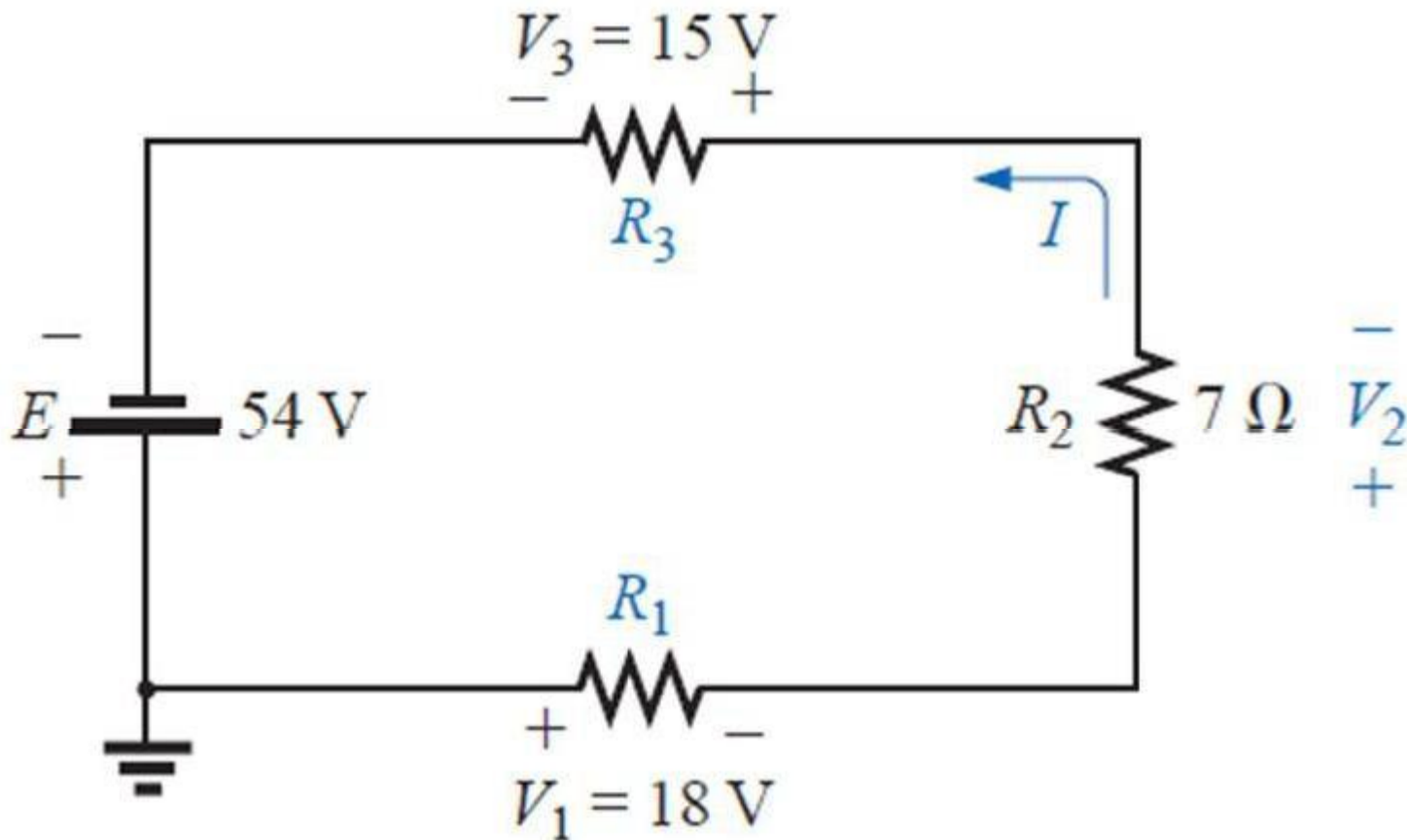
Example

2) For the following circuit:

a) Determine V_2 using Kirchhoff's voltage law.

b) Determine I .

c) Find R_1 and R_3 .

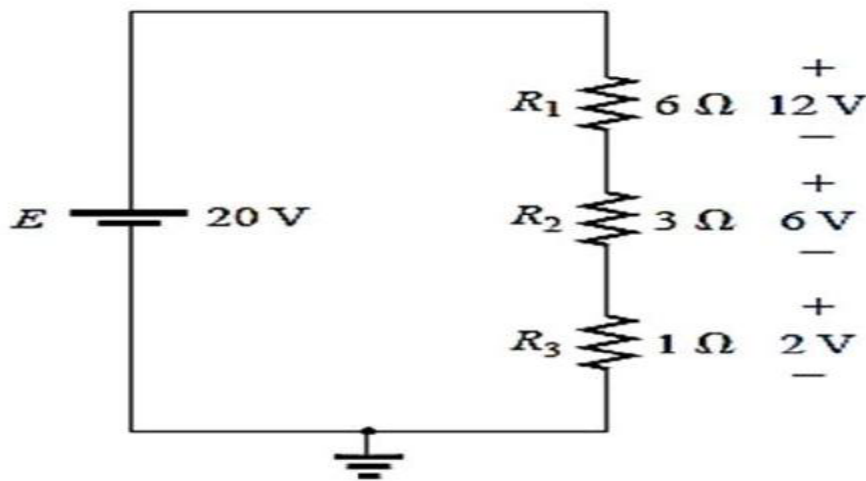


VOLTAGE DIVIDER RULE

The voltage dropped across any series resistor is proportional to the magnitude of the resistor. The total voltage dropped across all resistors must equal the applied voltage source(s) by KVL.

The voltage divider rule allows us to determine the voltage across any series resistance in a single step, without first calculating the current.

Consider the following circuit:



To derive the voltage divider formula, consider the following circuit;

and

$$R_T = R_1 + R_2$$

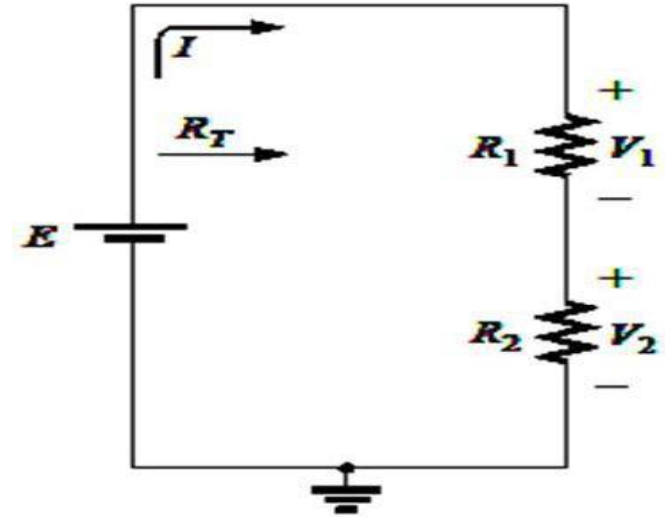
$$I = \frac{E}{R_T}$$

Applying Ohm's law:

$$V_1 = IR_1 = \left(\frac{E}{R_T}\right)R_1 = \frac{R_1 E}{R_T}$$

with

$$V_2 = IR_2 = \left(\frac{E}{R_T}\right)R_2 = \frac{R_2 E}{R_T}$$



In general, to find the voltage across R_x in series circuit is:

$$V_x = \frac{R_x E}{R_T}$$

Where: V_x : the voltage across R_x

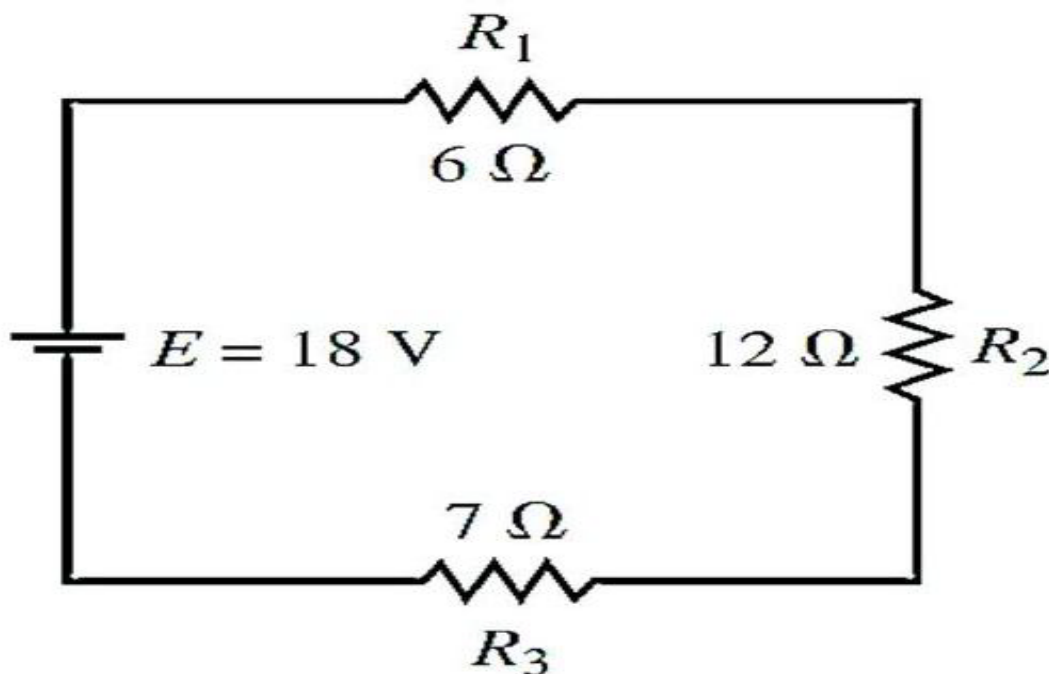
E : source voltage

R_t : Total resistance of series ckt

In words, the voltage divider rule states that the voltage across a resistor in a series circuit is equal to the value of that resistor times the total impressed voltage across the series elements divided by the total resistance of the series elements.

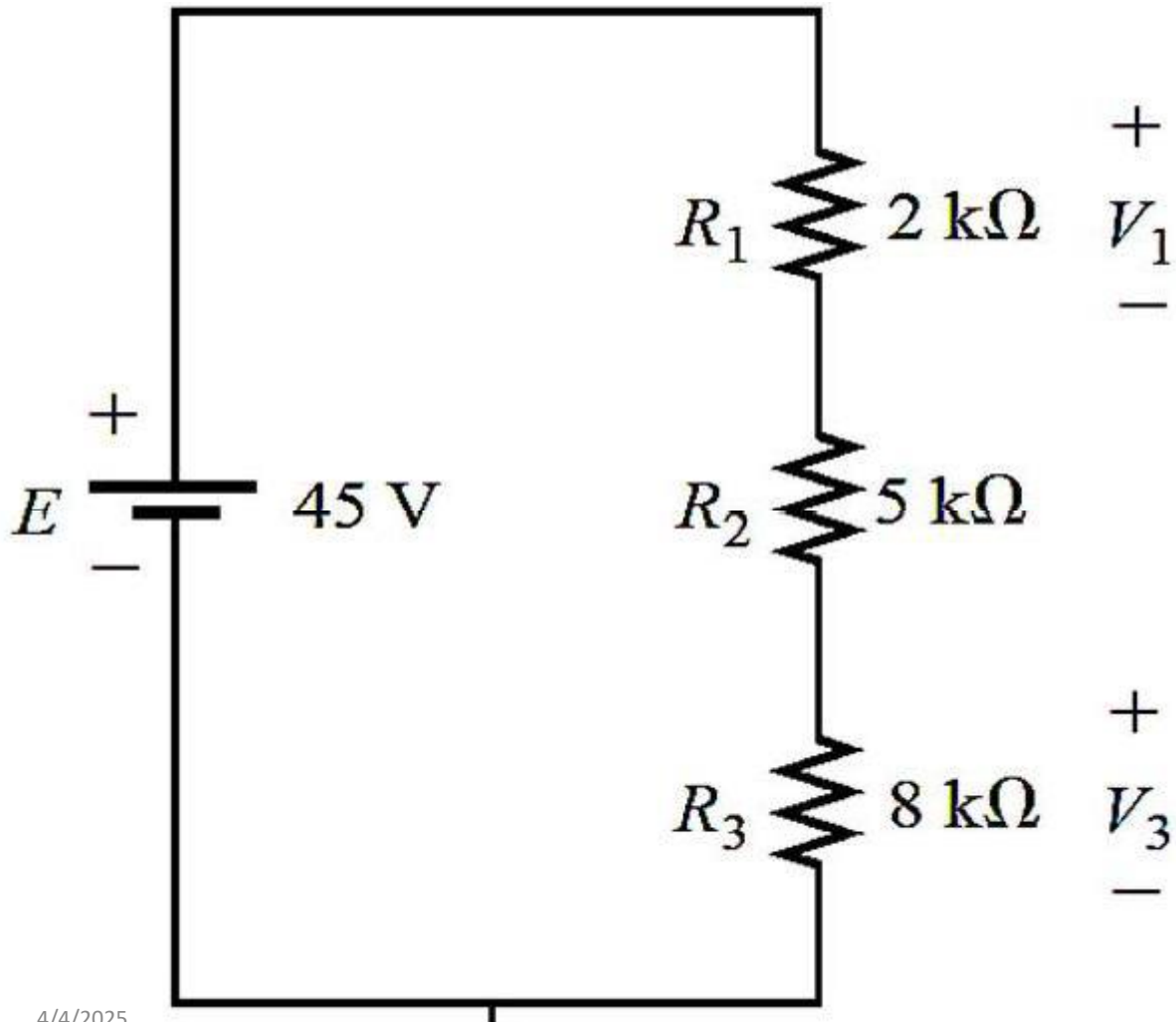
Example:

1) Use the voltage divider rule to determine the voltage across each of the resistors in the following circuit. Show that the summation of voltage drops is equal to the applied voltage rise in the circuit.



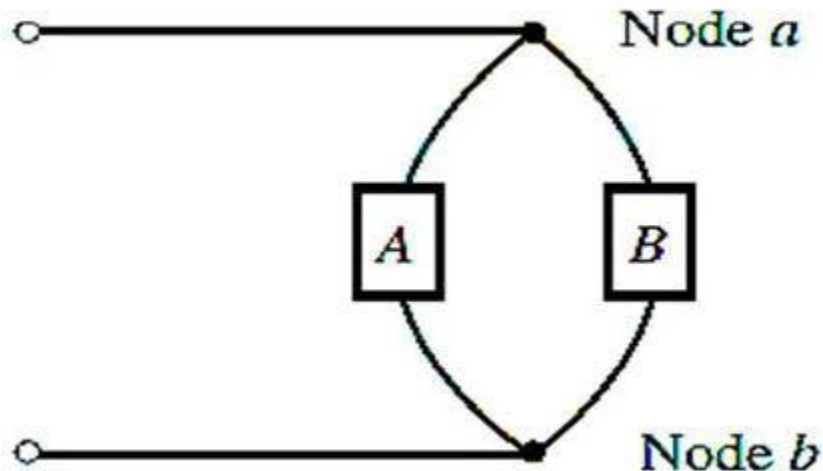
Example:

2) Using the voltage divider rule, determine the voltages V_1 and V_3 for the following series circuit.

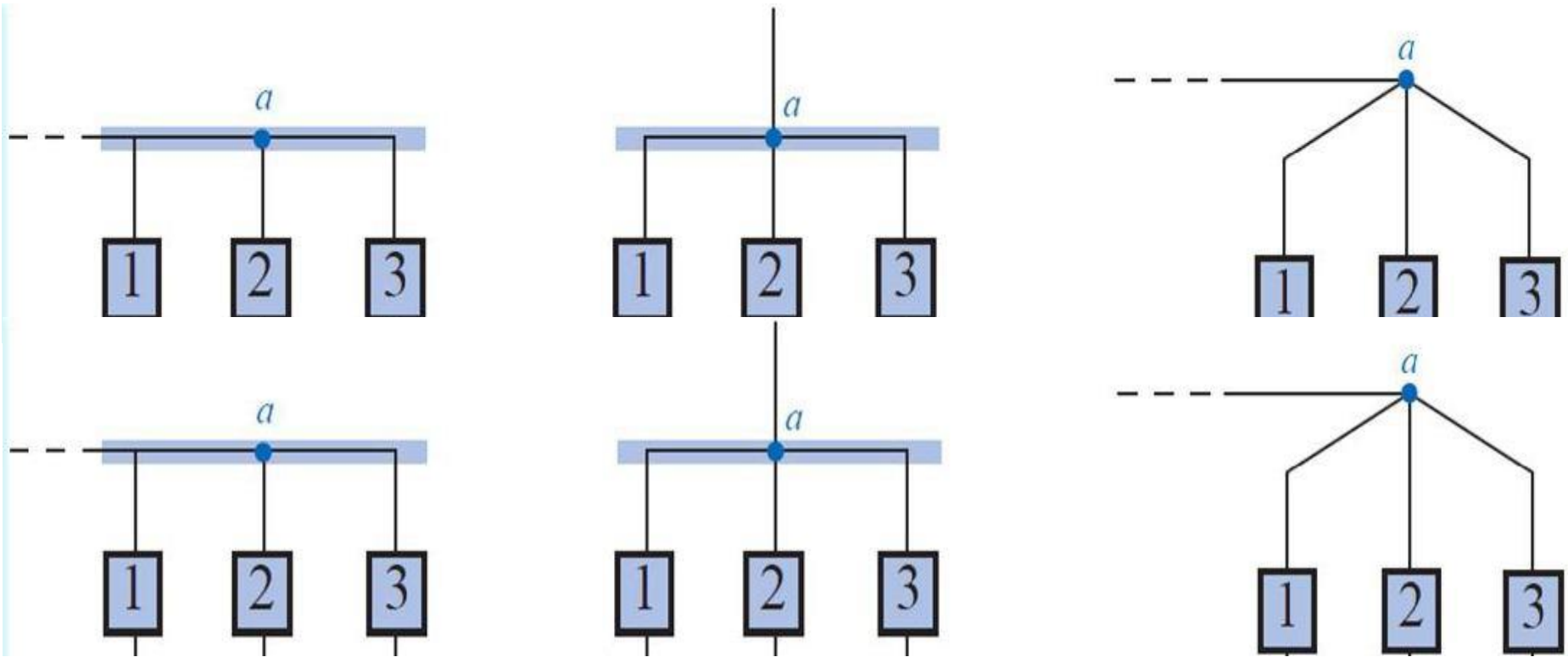


2. PARALLEL CIRCUIT

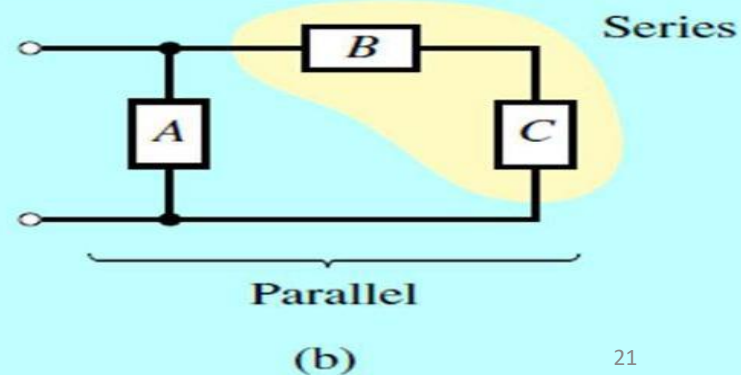
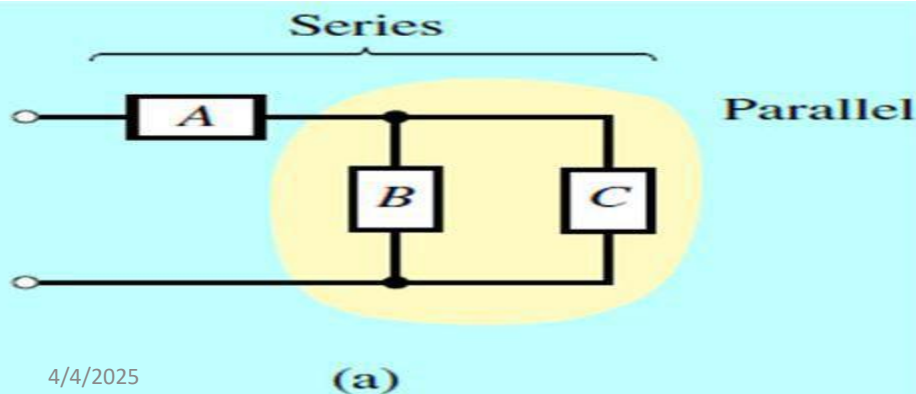
- We will now examine the parallel circuit and all the methods and laws associated with this important configuration.
- Two elements, branches, or networks are in parallel if they have exactly two points in common.
- These points of connection are often referred to as nodes. For instance, elements A & B are in parallel b/c they have terminals a & b in common.



- In Figure below, all the elements are in parallel because they satisfy the above criterion.
- Three configurations are provided to demonstrate how the parallel networks can be drawn.



- Very often, circuits contain a combination of series and parallel components. Consider the networks shown in Figure below.
- In the circuit of Figure (a), we see that element B is in parallel with element C since they each have common nodes. This parallel combination is now seen to be in series with element A.
- In the circuit of Figure (b), element B is in series with element C since these elements have a single common node.
- The branch consisting of the series combination of elements B and C is then determined to be in parallel with element A.



LAWS AND METHODS ASSOCIATED WITH PARALLEL CIRCUITS

KIRCHHOFF'S CURRENT LAW (KCL)

- Kirchhoff's current law states that the summation of currents entering a node is equal to the summation of currents leaving the node.

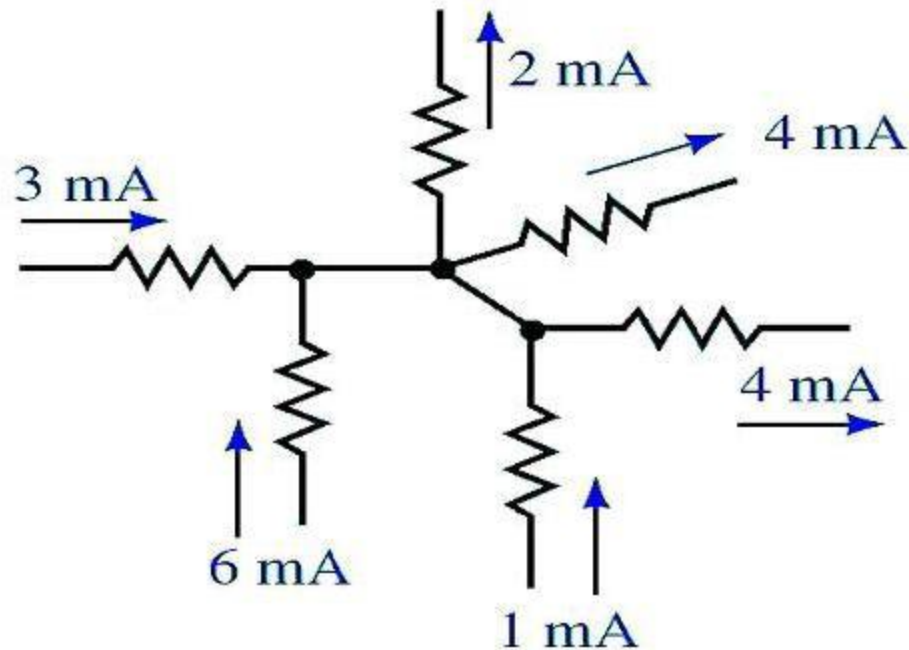
In mathematical form, Kirchhoff's current law is stated as follows:

$$\sum I_{\text{entering node}} = \sum I_{\text{leaving node}}$$

Example : Verify that Kirchhoff's current law applies at the node shown in figure below.

$$3\text{mA} + 6\text{mA} + 1\text{mA} = 2\text{mA} + 4\text{mA} + 4\text{mA}$$

$$10\text{mA} = 10\text{mA}$$



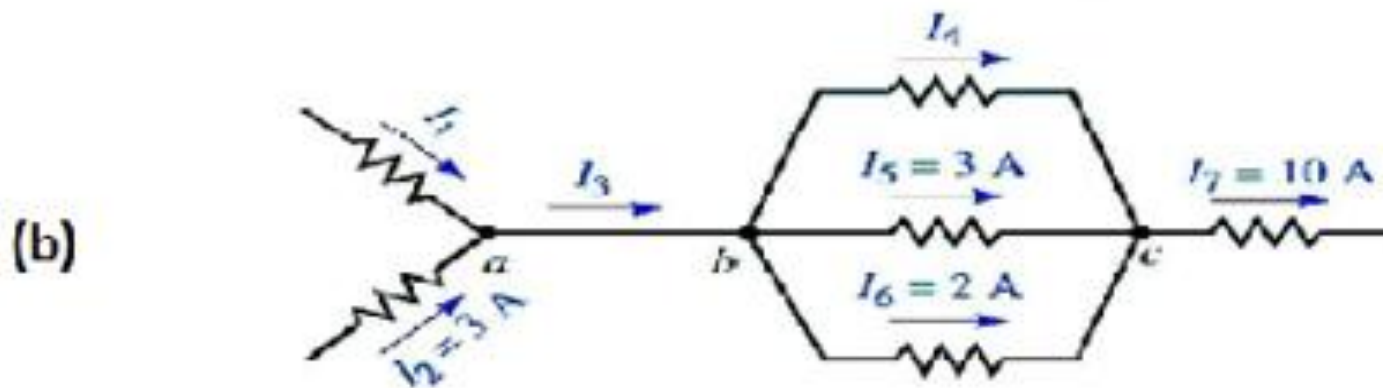
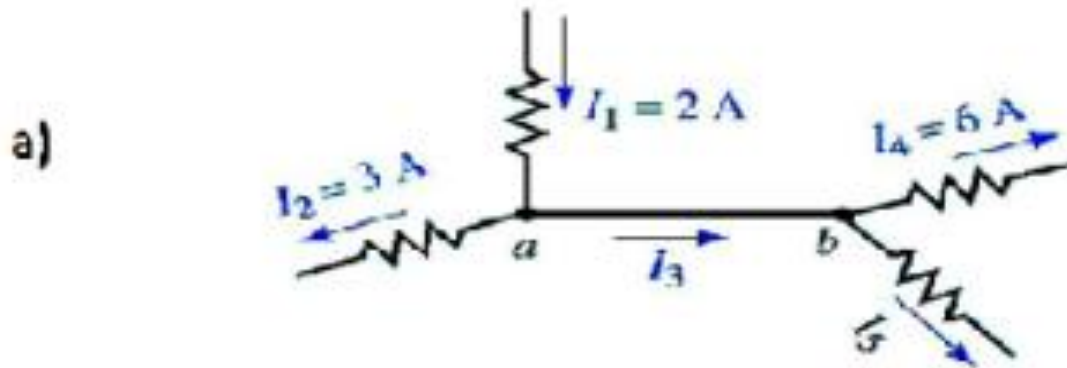
$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

$$3\text{mA} + 6\text{mA} + 1\text{mA} = 2\text{mA} + 4\text{mA} + 4\text{mA}$$

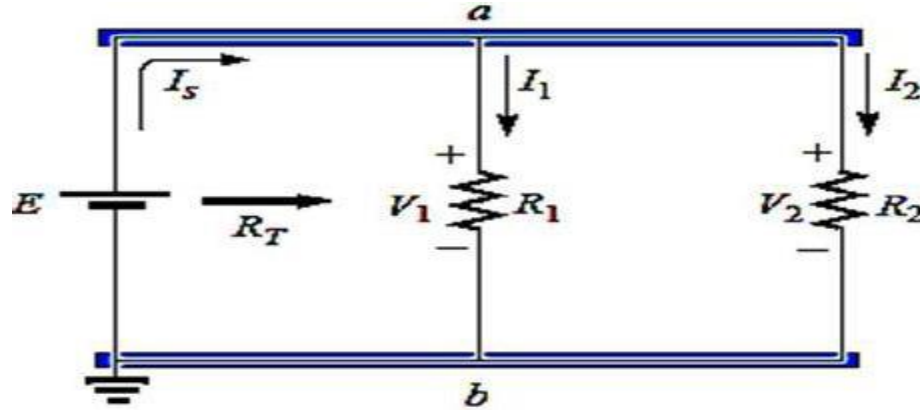
$$10\text{mA} = 10\text{mA}$$

Example :

Find the magnitudes of the unknown currents for the circuit of the following Figure (a) and (b) through applications of Kirchhoff's current law.



Let consider the following simplest parallel circuits.
All the elements have terminals a and b in common.



- Since the terminals of the battery are connected directly across the resistors R_1 and R_2 , the voltage across parallel elements is the same.
 $E = V_1 = V_2 = V_3 = \dots = V_n$
- If we were to apply Kirchhoff's voltage law around each closed loop in the parallel circuit of the above Figure, we would find that the voltage across all parallel resistors is exactly equal, ($E = V_1 = V_2$)

TOTAL RESISTANCE, R_T

➤ Applying Kirchhoff's current law to the above circuit , we have the expression: **$I_s = I_1 + I_2$**

➤ In general, for single-source parallel networks, the source current (I_s) is equal to the sum of the individual branch currents.

$$**$I_s = I_1 + I_2 + \dots + I_n$**$$

➤ Since E is the voltage across each of the parallel resistors, by Ohm's law, $I_1 = V_1/R_1$, $I_2 = V_2/R_2$ and so on.

By substitution into the equation above for current,

$$\frac{E}{R_T} = \frac{E}{R_1} + \frac{E}{R_2} + \dots + \frac{E}{R_n}$$

➤ Simplifying the above expression gives us the general expression for total resistance of a parallel circuit as

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

- Since conductance was defined as the reciprocal of resistance, we may write the above equation in terms of conductance, namely ,

$$G_T = G_1 + G_2 + \dots + G_n \quad (\text{S})$$

- An important effect of combining parallel resistors is that the resultant resistance will always be smaller than the smallest resistor in the combination.

For two resistors in parallel,

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

For three resistors in parallel

$$R_T = \frac{R_1 R_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3}$$

The power delivered by the source & that dissipated by the resistors is:

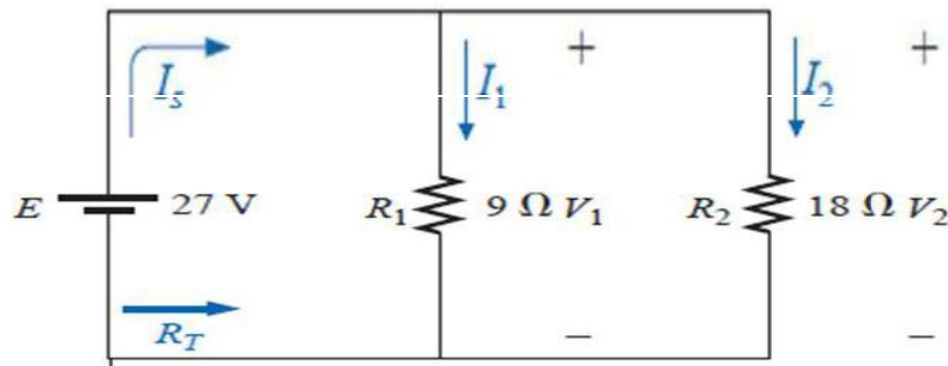
$$P_1 = V_1 I_1 = I_1^2 R_1 = \frac{V_1^2}{R_1}$$

$$P_2 = V_2 I_2 = I_2^2 R_2 = \frac{V_2^2}{R_2}$$

$$P_s = E I_s = I_s^2 R_T = \frac{E^2}{R_T}$$

EXAMPLE :-1) For the parallel network of the following circuit,

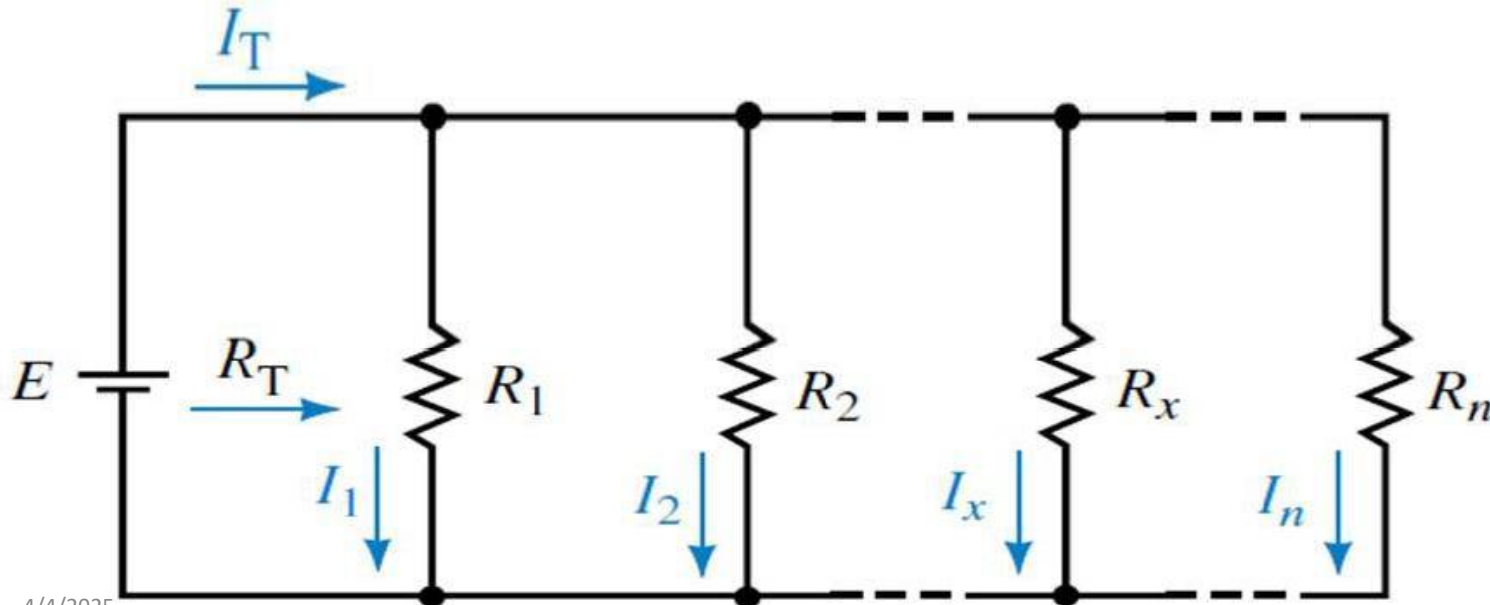
- Calculate R_T .
- Determine I_s .
- Calculate I_1 and I_2 , and demonstrate that $I_s = I_1 + I_2$.
- Determine the power to each resistive load.
- Determine the power delivered by the source, and compare it to the total power dissipated by the resistive elements.



CURRENT DIVIDER RULE

- In parallel networks, the voltage across all parallel elements is the same. However, the currents through the various elements are typically different.
- The current divider rule (CDR) is used to determine how current entering a node is split between the various parallel resistors connected to the node.

To drive the CDR, let Consider the network of parallel resistors shown in Figure below



- If this network of resistors is supplied by a voltage source, the total current in the circuit is

$$I_T = \frac{E}{R_T} \dots\dots\dots (*)$$

- Since each of the n parallel resistors has the same voltage, E, across its terminals, the current through any resistor in the network is given as.

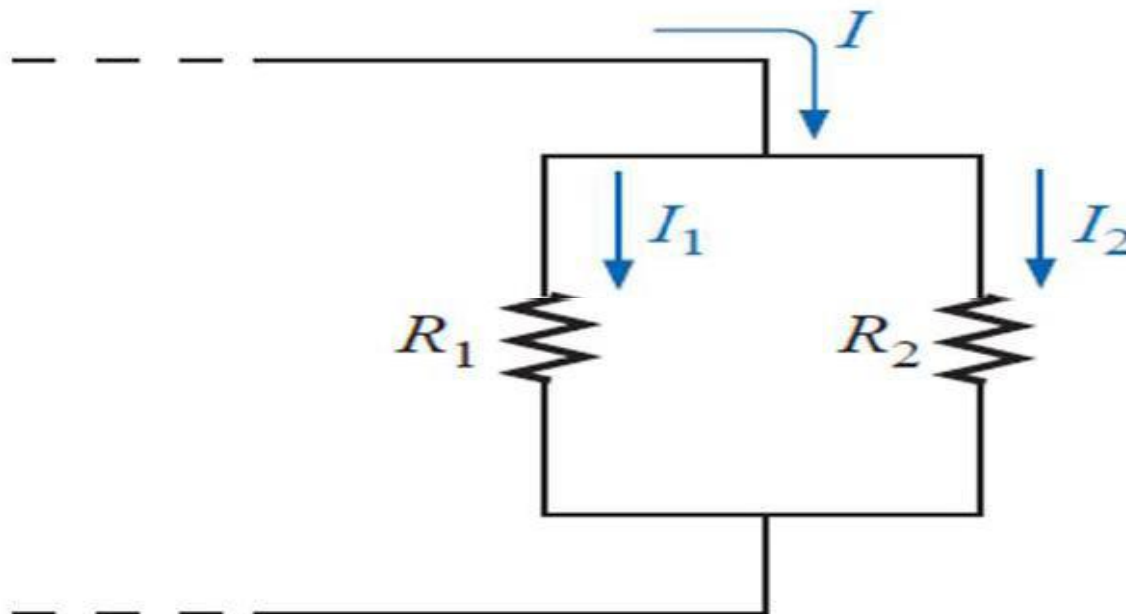
$$\dots I_x = \frac{E}{R_x} \dots\dots\dots (**)$$

- By rewriting Equation (*) as $E = I_T R_T$ and then substituting this into Equation (**), we obtain the general form for the current divider rule as follows:

$$I_x = \frac{R_T}{R_x} I_T$$

where I_x refers to the current through a parallel branch of resistance, R_x

- The CDR in words, the current through any parallel branch is equal to the product of the total resistance of the parallel branches and the input current divided by the resistance of the branch through which the current is to be determined.
- If the network consists of only two parallel resistors, then the current through each resistor may be found as below



➤ Recall that for two resistors in parallel, the total parallel resistance is given as

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$
$$I_1 = \frac{I_T R_T}{R_1} = \frac{I_T \left(\frac{R_1 R_2}{R_1 + R_2} \right)}{R_1}$$

Which simplifies,

$$I_1 = \frac{R_2}{R_1 + R_2} I_T$$

Similarly,

$$I_2 = \frac{R_1}{R_1 + R_2} I_T$$

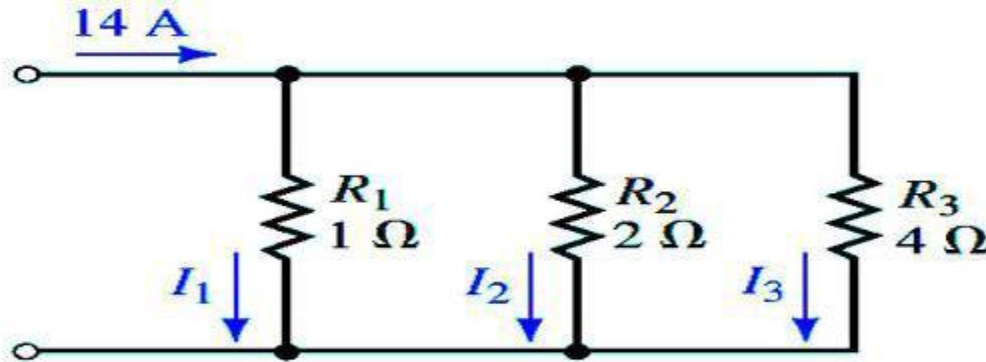
➤ In words, for two parallel branches, the current through either branch is equal to the product of the other parallel resistor and the input current divided by the sum (not the total parallel resistance) of the two parallel resistances.

Several other important characteristics of parallel networks become evident.

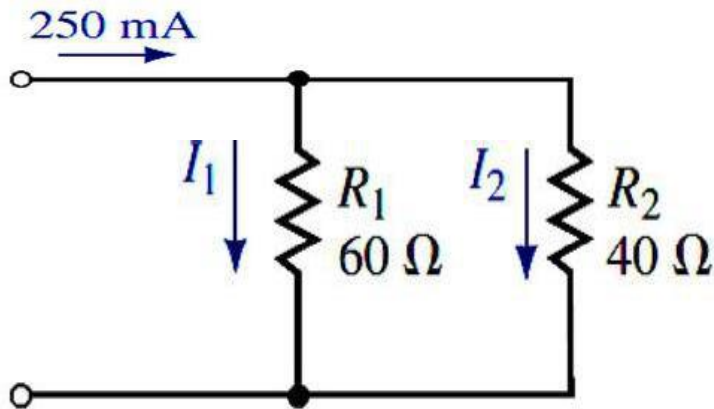
- ✓ If current enters a parallel network consisting of any number of equal resistors, then the current entering the network will split equally between all of the resistors.
- ✓ If current enters a parallel network consisting of several values of resistance, then the smallest value of resistor in the network will have the largest amount of current.
- ✓ Inversely, the largest value of resistance will have the smallest amount of current.
- ✓ This characteristic may be simplified by saying that most of the current will follow the path of least resistance.

EXAMPLE

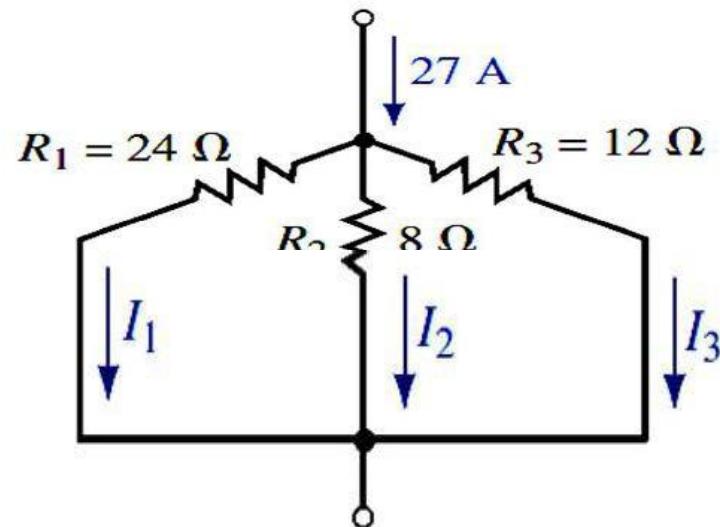
1) For the network of Figure below, determine currents I_1 , I_2 , and I_3



2) Use the current divider rule to calculate the unknown currents for the networks of Figure below.



(a) Network 1



(b) Network 2

3) SERIES -PARALLEL CIRCUITS

- In general, series-parallel networks are networks that contain both series and parallel circuit configurations
- By practicing (not memorizing) the techniques outlined in this section, you will find that most circuits can be reduced to groupings of series and parallel combinations.

In analyzing such circuits, it is imperative to remember that the rules for analyzing series and parallel elements still apply.

- The same current occurs through all series elements.
- The same voltage occurs across all parallel elements.
- In addition, remember that Kirchhoff's voltage law and Kirchhoff's current law apply for all circuits regardless of whether the circuits are series, parallel, or series-parallel

Example:- 1) Consider the circuit of Figure below

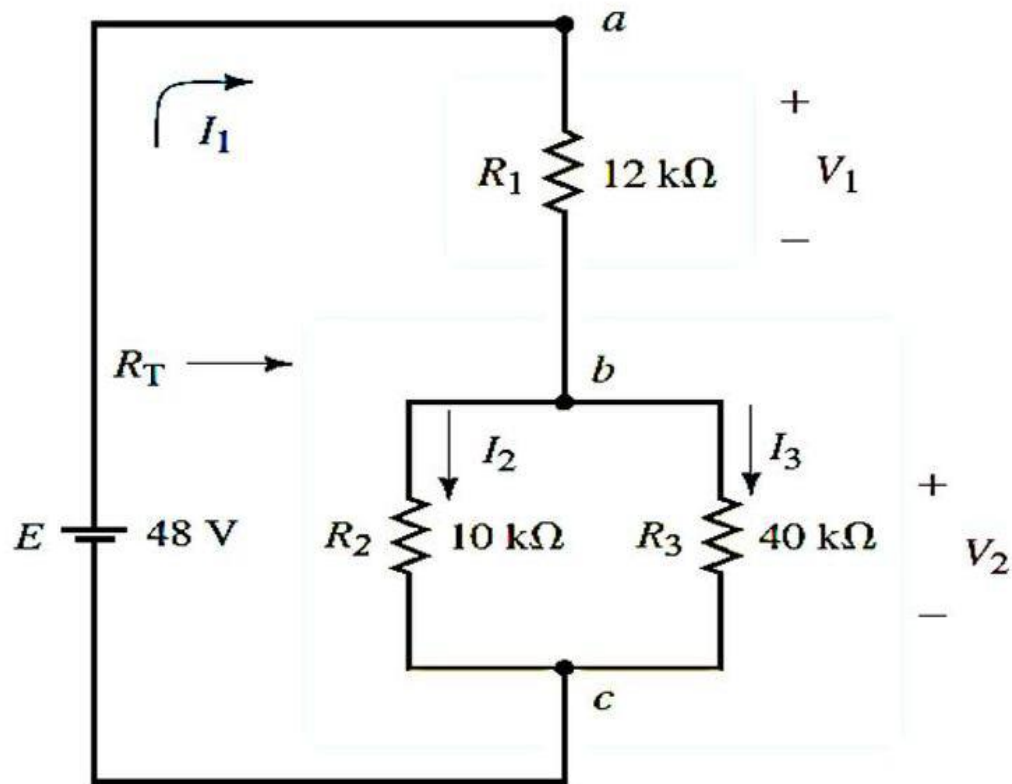
a) Find R_T

.

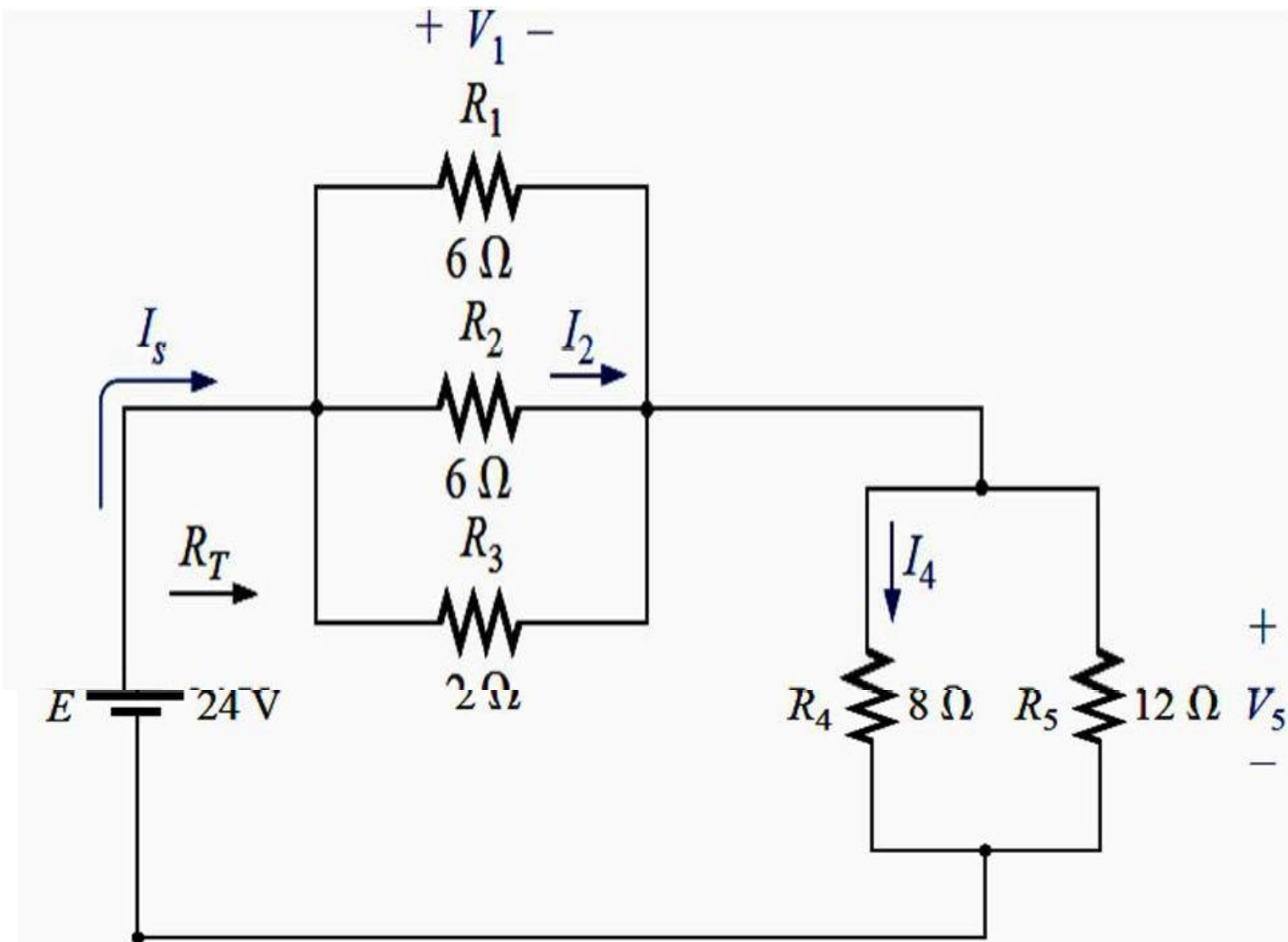
b) Calculate I_1 , I_2 , and I_3

.

c) Determine the voltages V_1 and V_2



Example:- 2) Find the indicated currents and voltages for the network of Figure below



4) COMPLEX CIRCUITS(Star-Delta circuits)

Complex circuits are circuits that contains none of the three types so that it is impossible to say parallel, series, or series-parallel.

Such kind of circuits arrangement is analyzed using the following two Kirchhoff's laws:

- Kirchhoff's Voltage law(KVL)

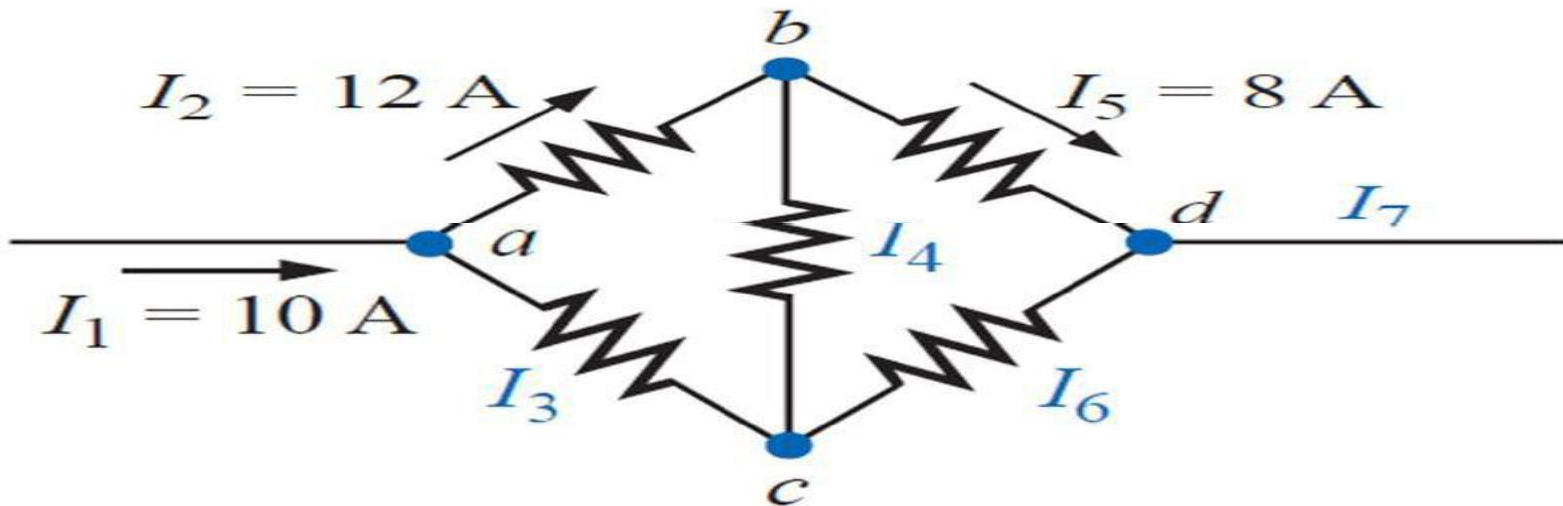
$$\sum_{\text{C}} V_{\text{rises}} = \sum_{\text{C}} V_{\text{drops}}$$

- Kirchhoff's Current law(KCL)

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

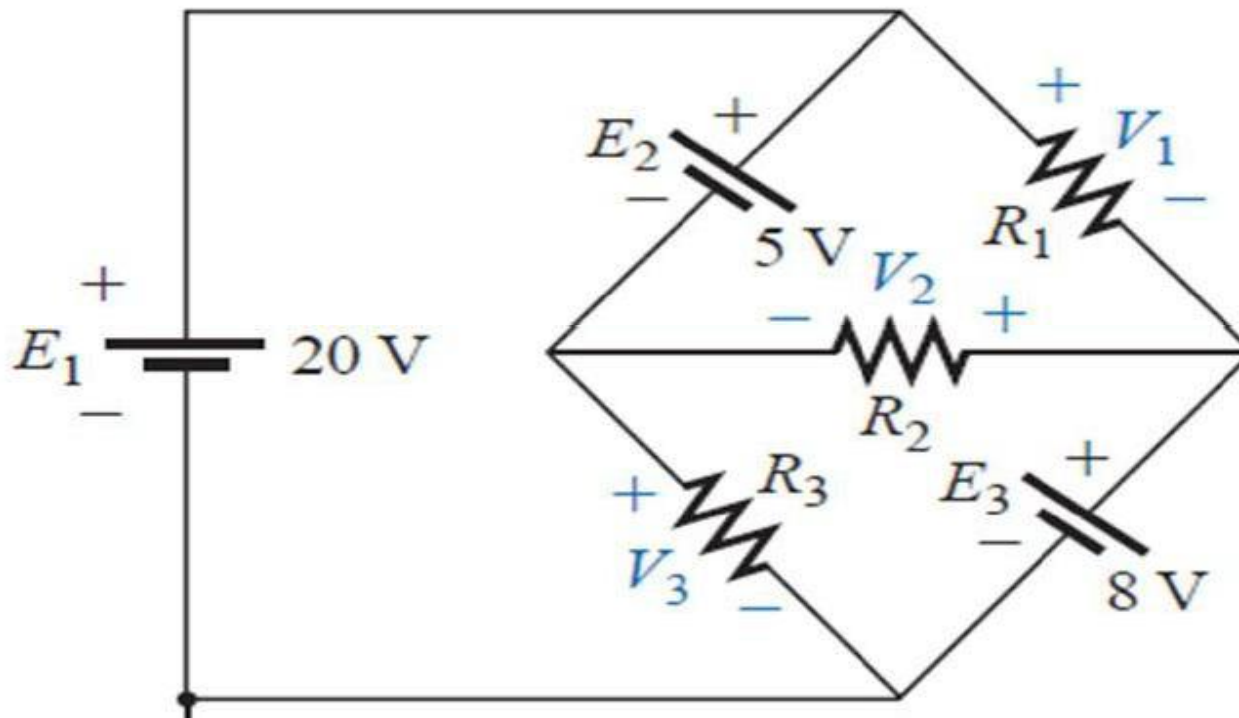
Example

Find the magnitude and direction of the currents I_3 , I_4 , I_6 , and I_7 for the network of Figure below. Even though the elements are not in series or parallel, Kirchhoff's current law can be applied to determine all the unknown currents.



Example

This example demonstrates the power of Kirchhoff's voltage law by determining the voltages V_1 , V_2 , and V_3 for the network of Figure below



To provide motivation for this topic, consider the circuit in Fig. 2.34. Note that this network has essentially the same number of elements as contained in our recent examples. However, when we attempt to reduce the circuit to an equivalent network containing the source V_1 and an equivalent resistor R , we find that nowhere is a resistor in series or parallel with another. Therefore, we cannot attack the problem directly using the techniques that we have learned thus far. We can, however, replace one portion of the network with an equivalent circuit, and this conversion will permit us, with ease, to reduce the combination of resistors to a single equivalent resistance. This conversion is called the wye-to-delta or delta-to-wye transformation.

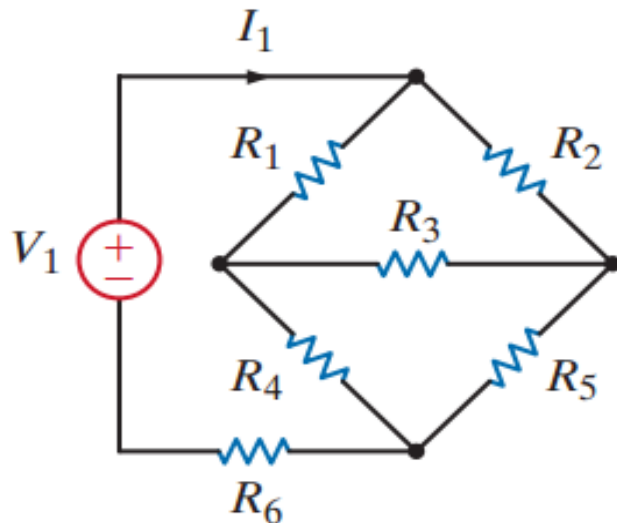


Figure 2.34

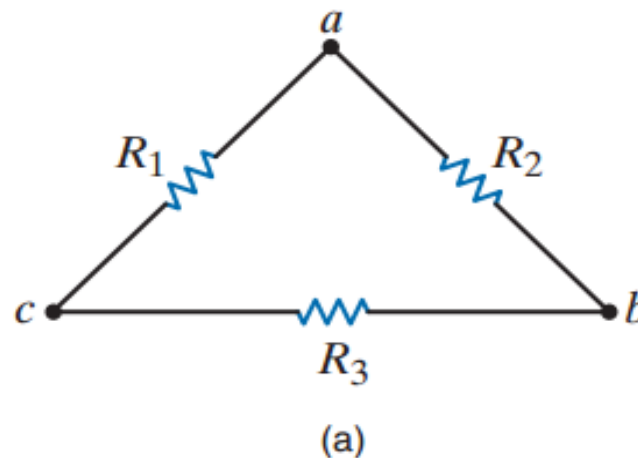
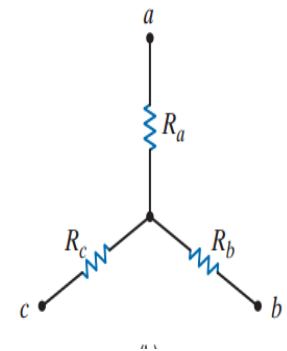


Figure 2.35



The transformation that relates the resistances R_1 , R_2 , and R_3 to the resistances R_a , R_b , and R_c is derived as follows. For the two networks to be equivalent at each corresponding pair of terminals, it is necessary that the resistance at the corresponding terminals be equal (e.g., the resistance at terminals a and b with c open-circuited must be the same for both networks).

Therefore, if we equate the resistances for each corresponding set of terminals, the following equations:

$$\begin{aligned} R_{ab} &= R_a + R_b = \frac{R_2(R_1 + R_3)}{R_2 + R_1 + R_3} \\ R_{bc} &= R_b + R_c = \frac{R_3(R_1 + R_2)}{R_3 + R_1 + R_2} \\ R_{ca} &= R_c + R_a = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3} \end{aligned}$$

Solving this set of equations for R_a , R_b , and R_c yields

$$\begin{aligned} R_a &= \frac{R_1 R_2}{R_1 + R_2 + R_3} \\ R_b &= \frac{R_2 R_3}{R_1 + R_2 + R_3} \\ R_c &= \frac{R_1 R_3}{R_1 + R_2 + R_3} \end{aligned}$$

Similarly, if we solve Eq. (2.27) for R_1 , R_2 , and R_3 , we obtain

$$\begin{aligned} R_1 &= \frac{R_a R_b + R_b R_c + R_a R_c}{R_b} \\ R_2 &= \frac{R_a R_b + R_b R_c + R_a R_c}{R_c} \end{aligned}$$

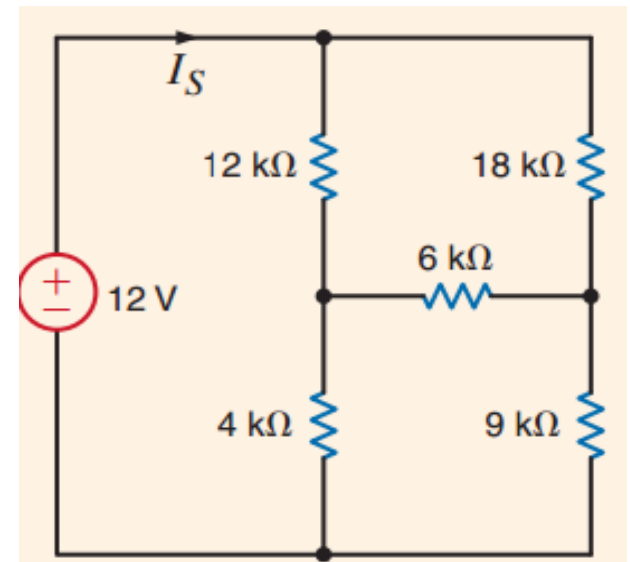
$$R_3 = \frac{R_a R_b + R_b R_c + R_a R_c}{R_a}$$

For balanced case where $R_a = R_b = R_c$ and $R_1 = R_2 = R_3$, the above equations reduced to:

$$R_Y = \frac{1}{3} R_\Delta$$

$$R_\Delta = 3R_Y$$

Example: For the given network find the source current I_S



Solution

Change the Delta connected to Wye as the following

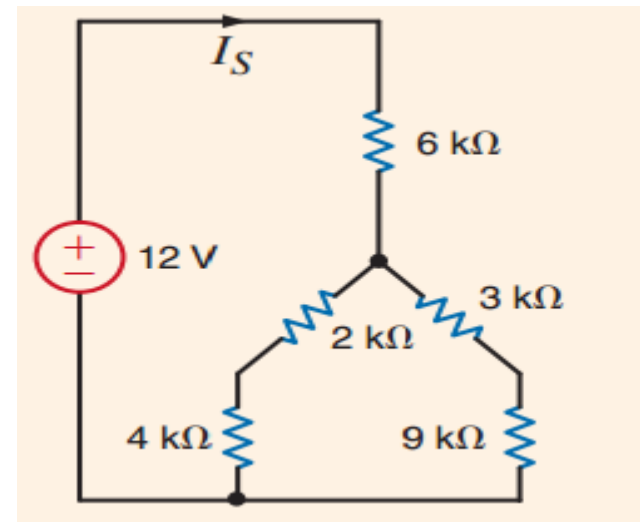
$$R_a = \frac{12K \cdot 18K}{12 + 6 + 18K} = 6K$$

$$R_b = \frac{18K \cdot 6K}{36K} = 3K$$

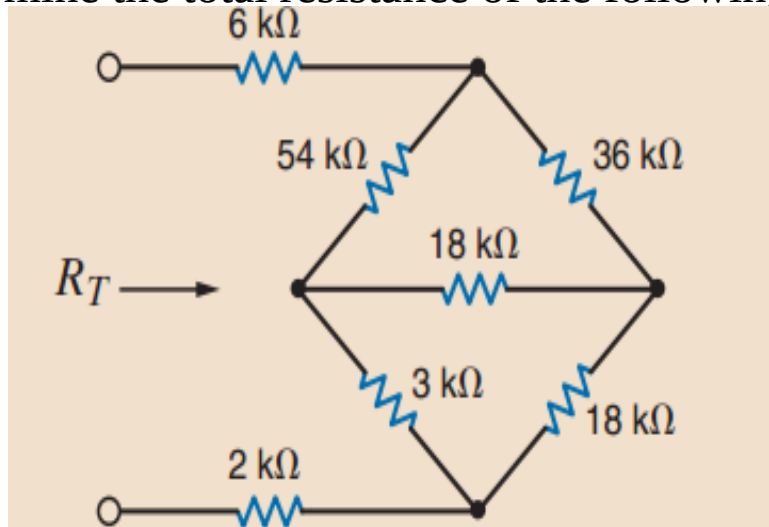
$$R_c = \frac{12K \cdot 6K}{36K} = 2K$$

$$R_T = 6K + 4K = 10K$$

$$\text{And } I_s = 12V / 10K = 1.2mA$$



Example 2 : Determine the total resistance of the following circuit



THANK YOU!!!